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RFC-ATF-6

Agent Trust Fabric -- Behavioral Execution Verification Protocol

Behavioral Anchor Record (BAR) -- Constraint Conformance Signal (CCS) -- Cross-Turn Coherence Hash Chain (CTCHC)

Harold Alberto Nunes Rodelo, Editor

OMNIX QUANTUM LTD -- 71-75 Shelton Street, Covent Garden, London WC2H 9JQ, England

standards@omnixquantum.com -- omnixquantum.net

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Extension to RFC-ATF-1, RFC-ATF-2, RFC-ATF-3, RFC-ATF-4, and RFC-ATF-5

Compliance Designation: ATF-BEV-Compliant -- Sixth and highest tier
18 new invariants (BEV-INV-001-018) -- 106 total ATF invariants -- 20 protocol families

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Abstract

RFC-ATF-6 specifies the **Behavioral Execution Verification Layer (BEV)** of the Agent Trust Fabric -- the sixth and final RFC in the ATF Open Standard series. RFC-ATF-6 answers the structural question that no prior RFC addresses: *did the agent actually behave within its authorized constraints during execution, turn by turn?*

The five prior RFCs establish governance infrastructure: who authorized (RFC-ATF-1), was authority continuously valid (RFC-ATF-2), where does evidence go (RFC-ATF-3), was monitoring active between requests (RFC-ATF-4), and what else could have happened (RFC-ATF-5). All five share a structural assumption: the authorized action is treated as a black box. The governance receipt authorizes. The execution receipt records. But no ATF artifact through RFC-ATF-5 contains a cryptographic record of what the agent actually produced -- the Behavioral Execution Gap (Gap_BEV).

RFC-ATF-6 closes this gap with three new protocol components: the **Behavioral Anchor Record (BAR)**, the **Constraint Conformance Signal (CCS)**, and the **Cross-Turn Coherence Hash Chain (CTCHC)**. These components are additive, backward compatible, and constitute Layer 6 of the ATF stack. Together they produce the first formally specified Session Integrity Proof (SIP) for multi-turn AI agent interactions -- independently verifiable offline using only the artifact bundle and the platform's Dilithium-3 (ML-DSA-65, FIPS 204) public key.

1. Problem Statement -- The Behavioral Execution Gap

1.1 Gap_BAG -- Behavioral Attestation Gap

Every prior ATF artifact treats the authorized action as a black box. The Governing Receipt (GR) authorizes the action. The Execution Receipt records that the action occurred. But no artifact cryptographically binds the GR to what the agent actually produced during execution -- the Behavioral Output (BO). This is Gap_BAG:

```
Gap_BAG = { (GR, BO) : no verifiable cryptographic binding exists between GR and BO }

Consequences:
(a) Regulatory gap: EU AI Act Art.9 requires monitoring that AI operates within defined
boundaries.
Gap_BAG leaves compliance assertion without behavioral evidence.
(b) Forensic gap: Auditor cannot reconstruct what the agent actually said or produced.
Governance receipts are authorization evidence, not behavioral evidence.
(c) Liability gap: Legal counsel cannot prove agent operated within authorized
constraints.
Receipt says "authorized" -- not "behaved within bounds of authorization."
(d) Audit gap: SOC 2 Type II CC7.2 monitoring requires per-event evidence.
No BAR = no per-turn behavioral audit trail.
```

1.2 Gap_COP -- Conformance Observability Problem

The AGVP watchdog (RFC-ATF-4) issues anticipatory vetoes based on observable conformance signals. For structural governance (CES, fragmentation) and semantic drift (DSPP), signals exist. For behavioral output conformance -- whether the agent's actual outputs satisfy the constraint set defined in the Governing Receipt -- no governance-native signal existed before this RFC. Gap_COP left the AGVP input space incomplete for behavioral trajectories:

```
Gap_COP: AGVP_INPUT_SPACE = {CES, fragmentation_index, DSPP_drift} --
behavioral_conformance

Without CCS:
AGVP can detect structural failures (authority over-delegation, fragmentation).
AGVP can detect semantic drift (token distribution shift).
AGVP CANNOT detect: agent outputs violating defined output domains.
AGVP CANNOT detect: agent responses breaching prohibited class constraints.
AGVP CANNOT detect: cumulative behavioral drift across multi-turn sessions.

With CCS (RFC-ATF-6):
AGVP_INPUT_SPACE includes conformance_score per turn.
Verdicts DRIFTING/BREACH/VIOLATION trigger PVR issuance (BEV-INV-007).
CCS completes AGVP input space for behavioral governance.
```

1.3 Gap_MCP -- Multi-Turn Coherence Problem

A set of valid BAR records does not prove Session Coherence without a chaining mechanism. An adversary could present a subset of BARs (omitting turns where violations occurred), reorder BARs from different sessions, or substitute BARs from prior compliant sessions to construct a fraudulent compliance report. Gap_MCP is the absence of a cryptographic Session Integrity Proof (SIP) that binds the complete turn sequence:

```
Gap_MCP attacks (without CTCHC):

(a) Omission attack: adversary presents {BAR_0, BAR_1, BAR_3} -- BAR_2 violated constraints

(b) Reordering attack: adversary presents {BAR_0, BAR_2, BAR_1} -- hides violation sequence

(c) Substitution attack: adversary replaces BAR_i with BAR from prior compliant session

(d) Session mixing: adversary interleaves BARs from two different session contexts

Without CTCHC: each BAR is independently valid -- no inter-BAR binding detects attacks (a)-(d).

With CTCHC (RFC-ATF-6):

link[n] = SHA3-256({prev: link[n-1], turn: output_hash_n, receipt: governing_receipt_id})

Attacks (a)-(d) all produce link[n] != CTCHC.final_chain_hash -> VERIFIED = FALSE.
```

1.4 Formal Definition: Gap_BEV

Gap_BEV = Gap_BAG ∪ Gap_COP ∪ Gap_MCP

Gap_BEV is the Behavioral Execution Gap -- the set of all governance properties that an ATF-compliant system through RFC-ATF-5 does not formally specify. RFC-ATF-6 closes Gap_BEV completely. The Behavioral Execution Verification Layer (BEV) is the set of protocol elements, record types, invariants, and formal proofs that constitute this closure.

2. Architecture Overview -- BEV Layer

2.1 ATF Stack -- Six-Layer Architecture

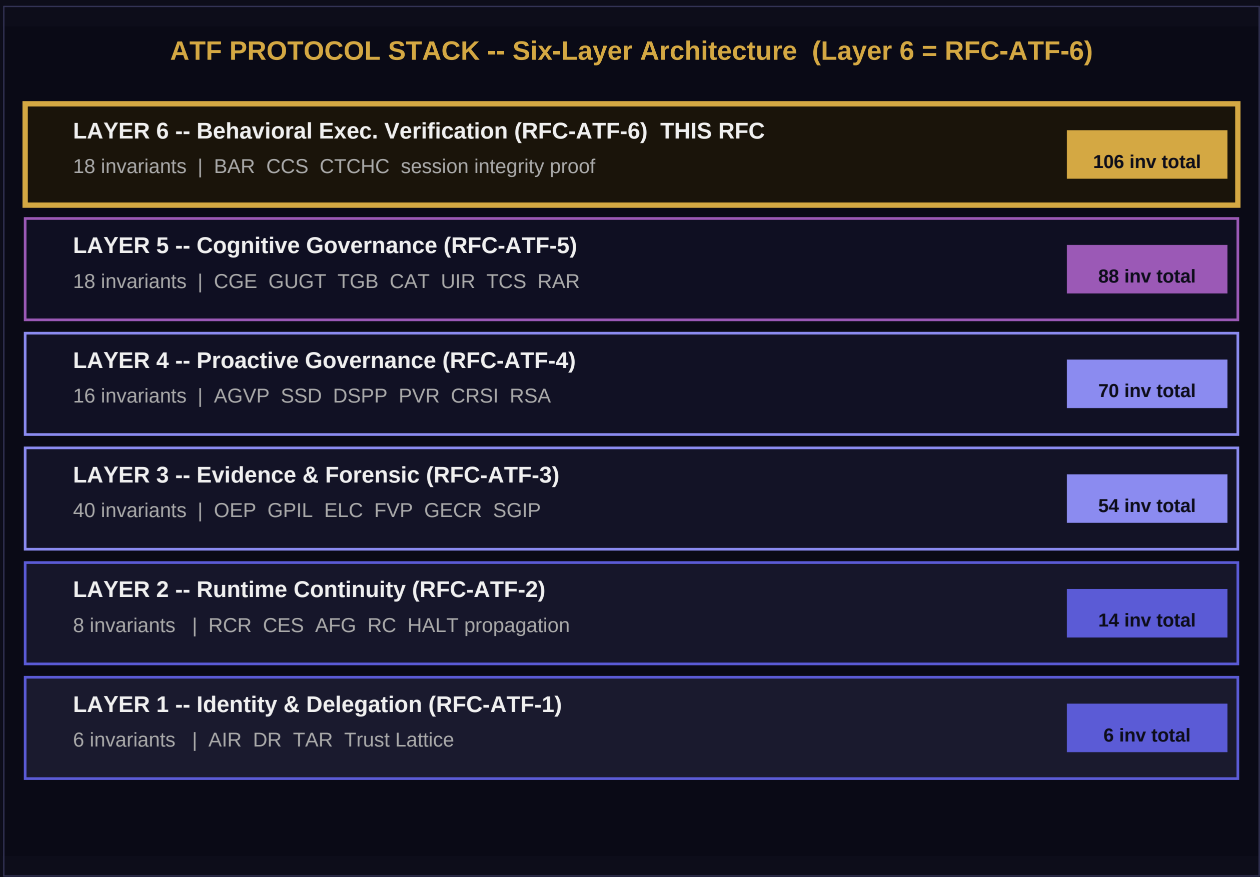

Figure 1. ATF six-layer protocol stack. Layer 6 (gold border) is introduced by RFC-ATF-6.

```
+==========================================================================+
| ATF PROTOCOL STACK -- SIX-LAYER COMPLETE ARCHITECTURE |
+==========================================================================+
| |
| Layer 6 -- BEHAVIORAL EXECUTION VERIFICATION LAYER (RFC-ATF-6) * THIS |
| +------------------------------------------------------------------------+
| | BAR: Behavioral Anchor Record |
| | -> PQC-signed per-turn attestation of actual agent output |
| | -> content_hash = SHA3-256(output_hash || receipt_id || turn_index) |
| | CCS: Constraint Conformance Signal |
| | -> governance-native conformance_score in [0.0, 1.0] per turn |
| | -> 4 components: OBS(40) + CSS(30) + SDS(20) + AAS(10) = 100 |
| | -> AGVP PVR triggered for verdict DRIFTING, BREACH, VIOLATION |
| | CTCHC: Cross-Turn Coherence Hash Chain |
| | -> genesis hash + per-turn links + ML-DSA-65 chain seal (SIP) |
| | -> offline-verifiable Session Integrity Proof |
| +------------------------------------------------------------------------+
| |
| Layer 5 -- COGNITIVE GOVERNANCE LAYER (RFC-ATF-5) |
```

```
| +----------------------------------------------------------------------+
| | CGE: Counterfactual Governance Engine (fragility_score) |
| | GUGT: Grand Unified Governance Theory (UIR, 6 UGIs) |
| | TGB: Temporal Governance Bridge (TCS, RAR, TMR) |
| +----------------------------------------------------------------------+
| |
| Layer 4 -- PROACTIVE GOVERNANCE PLANE (RFC-ATF-4) |
| +----------------------------------------------------------------------+
| | AGVP: Anticipatory Governance Veto Protocol (PVR) |
| | SSD: Structural Shift Detection (CRSI) |
| | DSPP: Dynamic Semantic Portability Protocol (RSA, SDR) |
| +----------------------------------------------------------------------+
| |
| Layer 3 -- EVIDENCE AND FORENSIC PLANE (RFC-ATF-3) |
| +----------------------------------------------------------------------+
| | GPIL: Governance Policy Interoperability Layer |
| | ELC: Evidence Lifecycle Classification (HOT/WARM/COLD) |
| | OEP: OMNIX Evidence Package -- offline-verifiable forensic bundle |
| | FVP: Forensic Verification Protocol |
| +----------------------------------------------------------------------+
| |
| Layer 2 -- RUNTIME CONTINUITY PLANE (RFC-ATF-2) |
| +----------------------------------------------------------------------+
| | RCR: Runtime Continuity Record |
| | CES: Coherence Evaluation Score (continuous health signal) |
| | AFG: Authority Fragmentation Guard |
| | RC: Reauthorization Challenge with HALT propagation |
| +----------------------------------------------------------------------+
| |
| Layer 1 -- IDENTITY AND DELEGATION PLANE (RFC-ATF-1) |
| +----------------------------------------------------------------------+
| | AIR: Agent Identity Record -- AID-{DOMAIN}-{16HEX} |
| | DR: Delegation Receipt -- authority budget chain |
| | TAR: Temporal Admissibility Record -- nanosecond admission proof |
| | Trust Lattice -- DAG of all delegation receipts |
| +----------------------------------------------------------------------+
+========================================================================+
| Invariant count: L1=6, L2=8, L3=40, L4=16, L5=18, L6=18 TOTAL: 106 |
| Compliance tier: ATF-BEV-Compliant (sixth and highest tier) |
+========================================================================+
```

2.2 BEV Module Independence and Failure Isolation

All three BEV modules are independently operable with fail-closed behavior. BAR is mandatory and blocks output delivery if unavailable. CCS and CTCHC are required for ATF-BEV-Compliant designation but individually configurable:

Module	Blocks output delivery?	Blocking on failure	Flag on failure
BAR	YES -- BEV-INV-001 (fail-closed)	Blocking -- no BAR, no output	SESSION_HALTED
CCS	NO -- non-blocking on failure	Non-blocking -- CCS_INCOMPLETE flag	CCS_INCOMPLETE on BAR
CTCHC	NO -- non-blocking on failure	Non-blocking -- CTCHC_INCOMPLETE flag	CTCHC_INCOMPLETE on session

2.3 BEV Record Sequencing per Turn

```
For every execution turn T_i of session S (BEV_ENABLED=true):

Step 1: Agent executes turn -- produces output_text (Behavioral Output, BO)
Step 2: output_hash = SHA3-256(output_text.encode('utf-8')) [BEV-INV-002]
Step 3: CCS computed from BO against Constraint Vector (CV) [BEV-INV-005]
Step 4: chain_link = SHA3-256(json.dumps({prev, turn, receipt})) [BEV-INV-011]
Step 5: BAR assembled (all fields populated including ccs_score) [BEV-INV-001]
Step 6: content_hash = SHA3-256(output_hash||receipt_id||str(turn_idx)) [BEV-INV-002]
Step 7: pqc_signature = dilithium3.sign(content_hash, secret_key) [BEV-INV-004]
Step 8: BAR persisted to atf_behavioral_anchor_records (append-only) [BEV-INV-004]
Step 9: If CCS.verdict in {DRIFTING, BREACH, VIOLATION}: AGVP notified [BEV-INV-007]
Step 10: If CCS.verdict == HALT (cumulative): session halted immediately [BEV-INV-008]

Ordering invariant: T(BAR.persist) < T(output.delivery) [BEV-INV-001]
Atomicity: BAR and CCS are produced as a single indivisible unit [BEV-INV-005]
```

3. Behavioral Anchor Record (BAR)

The BAR is the first ATF artifact that closes the authorization-to-output chain. It is a PQC-signed record produced at each execution turn, binding the SHA3-256 hash of the agent's actual output to the governing receipt and turn index. The three-input binding (output_hash || receipt_id || turn_index) makes the output, the authorization, and the position simultaneously tamper-evident.

3.1 BAR Architecture Flow

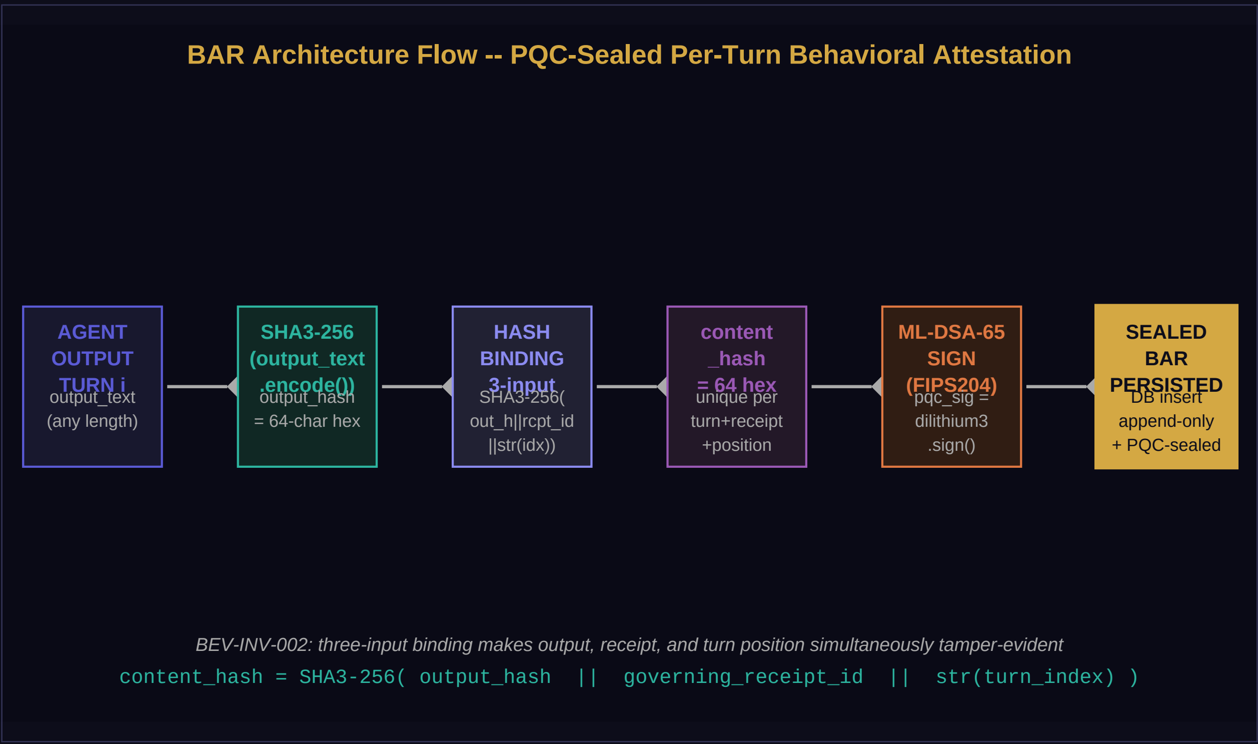

Figure 2. BAR creation pipeline. Every turn's output is hashed, bound to the governing receipt and turn index, then ML-DSA-65 signed before the BAR is persisted.

3.2 Content Hash Construction

```
# Step-by-step BAR content hash construction (BEV-INV-002)

output_hash = hashlib.sha3_256(output_text.encode('utf-8')).hexdigest()
# 64-char hex string -- SHA3-256 of raw UTF-8 output bytes

content_hash = hashlib.sha3_256(
(output_hash + governing_receipt_id + str(turn_index)).encode('utf-8')
).hexdigest()
# Binding: output + authorization receipt + position

pqc_signature = base64.b64encode(
dilithium3.sign(content_hash.encode('utf-8'), platform_secret_key)
).decode('utf-8')
# ML-DSA-65 (FIPS 204) over content_hash

pqc_algorithm = 'ML-DSA-65'

# Offline verification (any party, no network required):
```

```
assert dilithium3.verify(
content_hash.encode('utf-8'), pqc_signature_bytes, platform_public_key
)
```

3.3 BAR Full Specification

```
BAR (Behavioral Anchor Record) -- identifier: BAR-{HEX16}

COMMITTED FIELDS (included in content_hash):
bar_id : "BAR-4A2B8F1C3D5E7A9B" (BEV-INV-016)
session_id : "SESS-..."
governing_receipt_id : "ATFDR-..." (binding to GR)
agent_id : "AID-DOMAIN-..."
turn_index : 3 (0-based, strictly monotone)
output_hash : "a3f2b1c4d5e6f7a8..." (SHA3-256 of output_text)
output_hash_mode : "FULL" | "HASHED" | "REDACTED"
output_payload : { ...structured output... } (mode=FULL only)
constraint_vector : { ...CV from Governing Receipt... }
ccs_score : 87.50 (OBS+CSS+SDS+AAS)
ccs_verdict : "DRIFTING"
ccs_components : { obs:40, css:27.5, sds:18, aas:2 }
chain_link : "9f3a2b1c4d5e6f7a..." (CTCHC link for this turn)
genesis_hash : "..." (turn_index=0 only)
session_start_ns : 1748268000000000000 (turn_index=0 only)
bar_timestamp_ns : 1748268000123456789
issued_at : "2026-05-26T14:00:00.123456789+00:00"

SEAL FIELDS:
content_hash : SHA3-256(output_hash || governing_receipt_id || str(turn_index))
pqc_signature : base64(ML-DSA-65.sign(content_hash.encode()))
pqc_algorithm : "ML-DSA-65"
atf_spec_version : "RFC-ATF-6_v1.0"
bar_status : "ACTIVE" | "HALTED" | "VIOLATION"

METADATA:
created_at : TIMESTAMPTZ (DB server timestamp at INSERT)
```

3.4 Output Hash Privacy Modes

BAR supports three output payload modes to address GDPR, data minimization, and forensic requirements simultaneously. All modes produce a valid content_hash:

Mode	output_payload stored?	CCS computable?	GDPR default?	Use case
FULL	Yes -- complete output	Yes -- full CV check	No	Internal forensic audit, regulated outputs
HASHED	No -- hash only	Partial -- OBS+CSS only	Yes	Production default -- GDPR Art.5 minimization
REDACTED	No -- field omitted	No -- CCS_ENABLED=false	No	Privacy-sensitive regulated domains

3.5 BAR Invariants: BEV-INV-001-004, BEV-INV-015-016

Invariant	Statement	Violation Consequence
BEV-INV-001	Every execution turn MUST produce a BAR BEFORE output is delivered. No path delivers output without a BAR.	Session flagged BEV-INCOMPLETE. Session halted immediately.
BEV-INV-002	content_hash = SHA3-256(output_hash || governing_receipt_id || str(turn_index)). Three-input simultaneous binding.	BAR rejected as structurally invalid. Not accepted as attestation evidence.
BEV-INV-003	A BAR with bar_status=HALTED MUST immediately transition session to SESSION_HALTED. No subsequent turn permitted.	Critical governance violation -- session continues past HALT.
BEV-INV-004	Every BAR MUST be ML-DSA-65 signed over its content_hash. No UPDATE or DELETE on atf_behavioral_anchor_records.	BAR not independently verifiable. Append-only guarantee broken.
BEV-INV-015	output_text = "" or whitespace-only MUST set bar_status = VIOLATION. Silent outputs not permissible.	Governance gap -- agent produces no observable output, session continues.
BEV-INV-016	BAR identifier MUST conform to "BAR-{HEX16}": exactly 16 uppercase hexadecimal characters.	BAR not recognizable as canonical ATF artifact. Interoperability broken.

3.6 BAR Database Schema

```
-- Table: atf_behavioral_anchor_records (append-only, BEV-INV-004)
CREATE TABLE IF NOT EXISTS atf_behavioral_anchor_records (
bar_id VARCHAR(64) PRIMARY KEY, -- BAR-{HEX16}
session_id VARCHAR(64) NOT NULL,
governing_receipt_id VARCHAR(128) NOT NULL,
agent_id VARCHAR(128) NOT NULL,
turn_index INTEGER NOT NULL,
output_hash VARCHAR(80) NOT NULL, -- sha3_256:64hex
output_hash_mode VARCHAR(16) NOT NULL -- FULL|HASHED|REDACTED
CHECK (output_hash_mode IN ('FULL','HASHED','REDACTED')),
output_payload JSONB, -- mode=FULL only
constraint_vector JSONB NOT NULL,
ccs_score NUMERIC(7,4),
ccs_verdict VARCHAR(16),
ccs_components JSONB,
chain_link VARCHAR(64) NOT NULL,
genesis_hash VARCHAR(64),
session_start_ns BIGINT,
bar_timestamp_ns BIGINT NOT NULL,
issued_at TIMESTAMPTZ NOT NULL,
content_hash VARCHAR(64) NOT NULL,
pqc_signature TEXT NOT NULL,
pqc_algorithm VARCHAR(16) NOT NULL DEFAULT 'ML-DSA-65',
atf_spec_version VARCHAR(8) NOT NULL DEFAULT '1.6',
bar_status VARCHAR(16) NOT NULL DEFAULT 'ACTIVE'
```

```sql
CHECK (bar_status IN ('ACTIVE','HALTED','VIOLATION')),
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_bar_session ON atf_behavioral_anchor_records(session_id);
CREATE INDEX idx_bar_receipt ON atf_behavioral_anchor_records(governing_receipt_id);
CREATE INDEX idx_bar_turn ON atf_behavioral_anchor_records(session_id, turn_index);
CREATE INDEX idx_bar_status ON atf_behavioral_anchor_records(bar_status)
WHERE bar_status != 'ACTIVE';
```

4. Constraint Conformance Signal (CCS)

The CCS provides a governance-native behavioral conformance measurement for each execution turn. "Governance-native" means it references the actual Constraint Vector (CV) from the PQC-signed Governing Receipt -- the receipt IS the measurement specification. This is architecturally distinct from existing monitoring solutions (Arize, WhyLabs, Evidently) which monitor against separately configured policy files, creating a receipt-to-policy binding gap.

4.1 CCS Score Components and Verdict Ladder

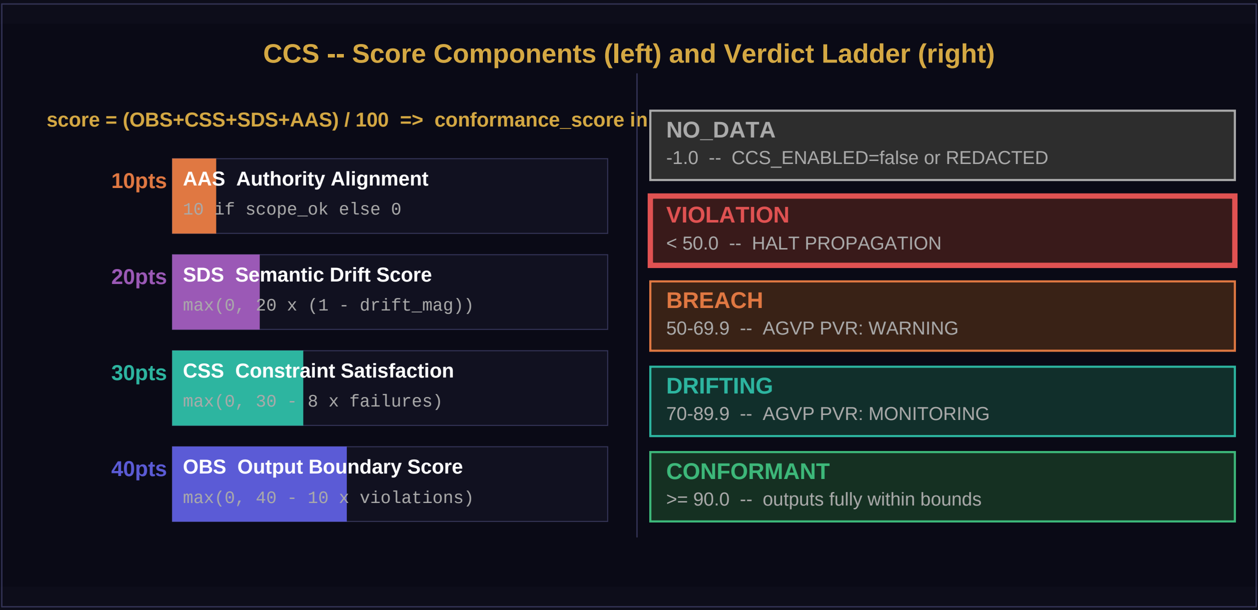

Figure 3. CCS score breakdown (left) and verdict ladder (right). Four components sum to 100 points, normalized to conformance_score in [0.0, 1.0] (BEV-INV-006).

4.2 Score Formula

```
# CCS component formulas (BEV-INV-006 requires all components >= 0)

OBS (Output Boundary Score, max=40):
obs_score = max(0.0, 40.0 - 10.0 * boundary_violation_count)
boundary_violation = output crosses a defined output domain boundary or prohibited class

CSS (Constraint Satisfaction Score, max=30):
css_score = max(0.0, 30.0 - 8.0 * constraint_failure_count)
constraint_failure = explicit constraint in CV.constraints[] evaluated to False

SDS (Semantic Drift Score, max=20):
sds_score = max(0.0, 20.0 * (1.0 - drift_magnitude))
drift_magnitude = cosine_distance(output_embedding, authorized_profile_embedding) in [0,
1]

AAS (Authority Alignment Score, max=10):
aas_score = 10.0 if authority_scope_satisfied else 0.0
authority_scope = capability/resource constraints from governing_receipt_id

# Aggregation
ccs_score = obs_score + css_score + sds_score + aas_score # range [0, 100]
```

```
conformance_score = ccs_score / 100.0 # range [0.0, 1.0]
drift_delta = 1.0 - conformance_score # per-turn drift
cumulative_drift += drift_delta # session accumulator

# Verdict assignment (thresholds configurable; defaults shown)
if conformance_score >= 0.90: verdict = "CONFORMANT"
elif conformance_score >= 0.70: verdict = "DRIFTING" # -> AGVP PVR: MONITORING
elif conformance_score >= 0.50: verdict = "BREACH" # -> AGVP PVR: WARNING
else: verdict = "VIOLATION" # -> HALT PROPAGATION

if cumulative_drift > CCS_DRIFT_THRESHOLD: # default 0.35, max 0.50
verdict = "HALT" # immediate session halt
```

4.3 CCS-AGVP Integration Loop

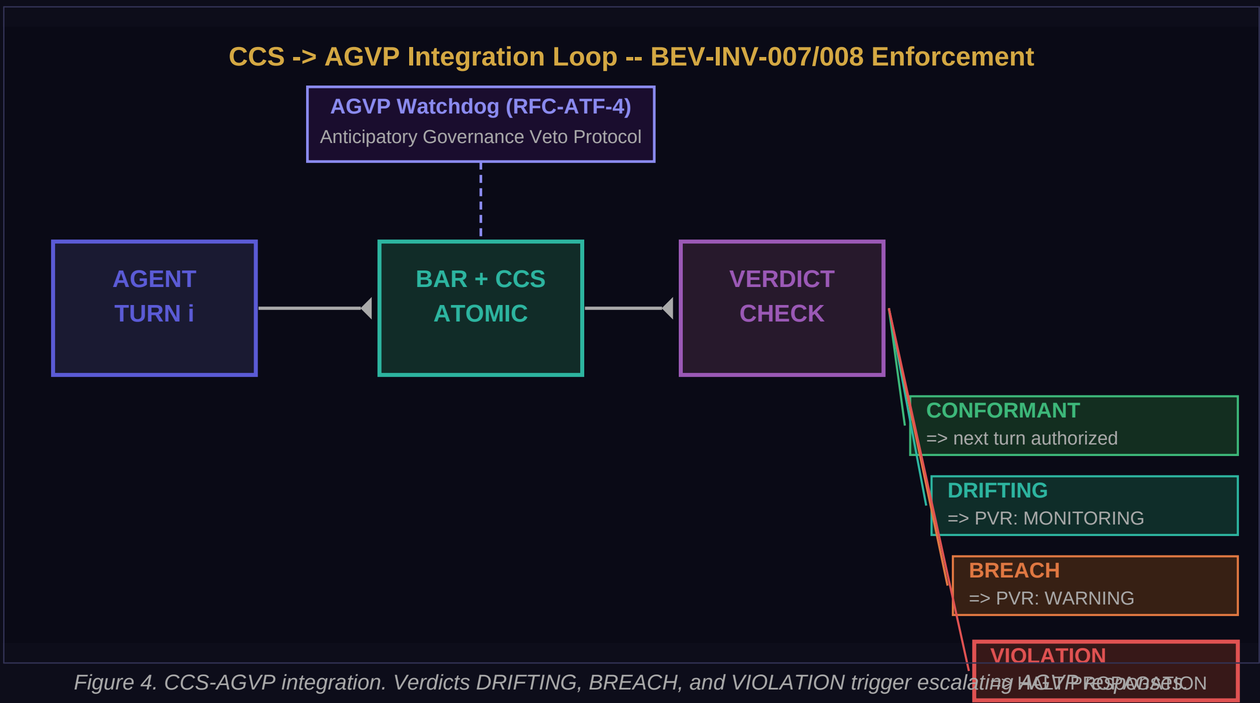

Figure 4. CCS-AGVP integration. Verdicts DRIFTING, BREACH, and VIOLATION trigger escalating AGVP responses. VIOLATION and HALT terminate the session before further outputs accumulate.

4.4 CCS Full Specification

```
CCS (Constraint Conformance Signal) -- identifier: CCS-{HEX16}

COMMITTED FIELDS:
ccs_id : "CCS-7F3A2B1C4D5E6F7A"
bar_id : "BAR-..." (FK -> BAR of this turn)
session_id : "SESS-..."
governing_receipt_id : "ATFDR-..."
turn_index : 3
conformance_score : 0.8750 in [0.0, 1.0]
verdict : "DRIFTING"
output_boundary_score : 40.00
constraint_satisfaction_score : 27.50
semantic_drift_score : 18.00
```

```
authority_alignment_score : 2.00
boundary_violation_count : 0
constraint_failure_count : 1
drift_magnitude : 0.100
cumulative_drift : 0.215 (session running total)
drift_delta : 0.125 (1.0 - conformance_score)
watchdog_triggered : true (verdict >= DRIFTING)
agvp_pvr_id : "PVR-..." (populated if PVR issued)
halt_triggered : false
chain_link_hash : "9f3a2b..." (CTCHC link at this turn)
computed_at_ns : 1748268000456789012
```

4.5 CCS Invariants: BEV-INV-005-009, BEV-INV-017

Invariant	Statement	Violation Consequence
BEV-INV-005	BAR and CCS MUST be produced in the same atomic operation. No valid BAR without CCS when CCS_ENABLED.	BAR without CCS fails ATF-BEV-Compliant. AGVP input space incomplete.
BEV-INV-006	conformance_score in [0.0, 1.0]. drift_delta = 1.0 - conformance_score. All four component scores >= 0.	Score out of bounds = score inflation attack vector. Verdict derived from invalid value is void.
BEV-INV-007	verdict in {DRIFTING, BREACH, VIOLATION} MUST set watchdog_triggered=true and trigger AGVP PVR.	Incomplete AGVP integration. ATF-BEV-Compliant certification blocked.
BEV-INV-008	cumulative_drift > CCS_DRIFT_THRESHOLD (default 0.35, max 0.50) MUST set verdict=HALT and halt session.	Session continues with excessive behavioral drift. Integrity failure.
BEV-INV-009	CCS records are append-only, ordered by turn_index. No UPDATE or DELETE on atf_constraint_conformance_signals.	Retroactive conformance falsification possible. Forensic integrity broken.
BEV-INV-017	cumulative_drift accumulator is isolated per session_id. On process restart, MUST reload from last persisted CCS.	Cross-session drift contamination. HALT triggered by unrelated session drift.

4.6 CCS Database Schema

```
-- Table: atf_constraint_conformance_signals (append-only, BEV-INV-009)
CREATE TABLE IF NOT EXISTS atf_constraint_conformance_signals (
ccs_id VARCHAR(64) PRIMARY KEY, -- CCS-{HEX16}
bar_id VARCHAR(64) REFERENCES atf_behavioral_anchor_records(bar_id),
session_id VARCHAR(64) NOT NULL,
governing_receipt_id VARCHAR(128) NOT NULL,
turn_index INTEGER NOT NULL,
conformance_score NUMERIC(7,4) NOT NULL
CHECK (conformance_score >= 0.0 AND conformance_score <= 1.0),
verdict VARCHAR(16) NOT NULL
CHECK (verdict IN ('CONFORMANT','DRIFTING','BREACH','VIOLATION','HALT','NO_DATA')),
```

```
output_boundary_score NUMERIC(6,3),
constraint_satisfaction_score NUMERIC(6,3),
semantic_drift_score NUMERIC(6,3),
authority_alignment_score NUMERIC(6,3),
boundary_violation_count INTEGER DEFAULT 0,
constraint_failure_count INTEGER DEFAULT 0,
drift_magnitude NUMERIC(7,4),
cumulative_drift NUMERIC(7,4) NOT NULL,
drift_delta NUMERIC(7,4) NOT NULL,
watchdog_triggered BOOLEAN NOT NULL DEFAULT FALSE,
agvp_pvr_id VARCHAR(64),
halt_triggered BOOLEAN NOT NULL DEFAULT FALSE,
chain_link_hash VARCHAR(64),
computed_at_ns BIGINT NOT NULL,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

5. Cross-Turn Coherence Hash Chain (CTCHC)

The CTCHC provides session-level behavioral integrity for multi-turn agent sessions. It chains every BAR's output hash with the prior link and governing receipt ID, seeding from a genesis hash bound to the session identity. The sealed CTCHC is the Session Integrity Proof (SIP) -- offline verifiable without any OMNIX infrastructure, resistant to omission, reordering, substitution, and session-mixing attacks.

5.1 CTCHC Chain Architecture

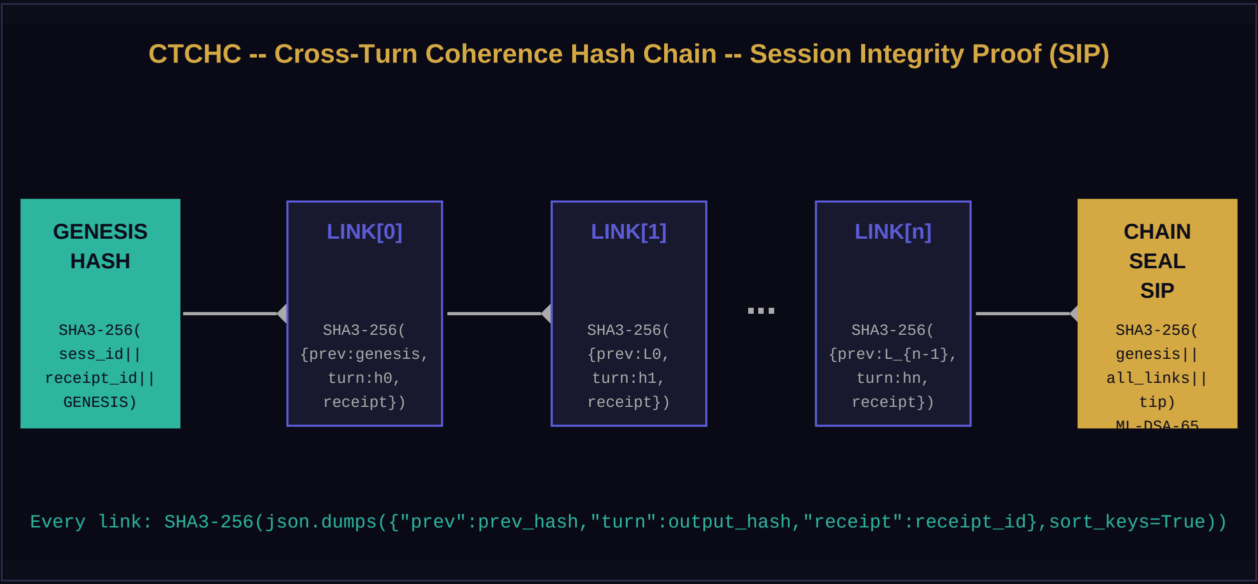

Figure 5. CTCHC hash chain. Genesis hash binds to session_id and governing_receipt_id. Each link encodes prior link, turn output hash, and receipt. ML-DSA-65 chain seal = SIP.

5.2 Chain Construction Algorithm

```
# CTCHC construction (BEV-INV-010 through BEV-INV-014, BEV-INV-018)

# Step 1: Genesis (BEFORE BAR_0 is produced -- BEV-INV-010)
genesis_hash = hashlib.sha3_256(
(session_id + '||' + governing_receipt_id + '||' + 'OMNIX-CTCHC-GENESIS').encode()
).hexdigest()

# Step 2: Per-turn link (n >= 0)
def compute_link(prev_hash, output_text, governing_receipt_id):
output_hash = hashlib.sha3_256(output_text.encode('utf-8')).hexdigest()
link_input = json.dumps(
{'prev': prev_hash, 'turn': output_hash, 'receipt': governing_receipt_id},
sort_keys=True # deterministic canonical form
).encode('utf-8')
return hashlib.sha3_256(link_input).hexdigest()

link[0] = compute_link(genesis_hash, output_0, governing_receipt_id)
link[n] = compute_link(link[n-1], output_n, governing_receipt_id)

# Step 3: Chain seal at session close (all N links must be present -- BEV-INV-012)
all_links_concat = ''.join(all_link_hashes) # ordered by turn_index
seal_hash = hashlib.sha3_256(
```

```
(genesis_hash + all_links_concat + link[N-1]).encode()
).hexdigest()

# Step 4: ML-DSA-65 seal (BEV-INV-014)
ctchc_seal = base64.b64encode(
dilithium3.sign(seal_hash.encode('utf-8'), platform_secret_key)
).decode('utf-8')

# BEV-INV-018: every link MUST carry same governing_receipt_id as genesis
assert all(link.governing_receipt_id == genesis_receipt_id for link in all_links)
```

5.3 Offline Verification Procedure (6 steps)

```
CTCHC OFFLINE VERIFICATION PROCEDURE (BEV-INV-012 + BEV-INV-013 + BEV-INV-014):

Input: ctchc.json, bar_list.json (ordered by turn_index), public_key.pem

Step 1 -- Verify CTCHC seal integrity
hash_check = sha3_256((ctchc.genesis_hash + all_link_hashes_concat +
ctchc.final_chain_hash).encode())
assert hash_check == ctchc.seal_hash # BEV-INV-013
verify ML-DSA-65(ctchc.ctchc_seal, ctchc.seal_hash, pubkey) # BEV-INV-014

Step 2 -- Verify turn completeness (BEV-INV-012)
assert {bar.turn_index for bar in bar_list} == set(range(len(bar_list)))
assert len(bar_list) == ctchc.turn_count
# Gaps in turn_index -> CTCHC_VERIFICATION_FAILED (omission attack detected)

Step 3 -- Verify each BAR individually
for bar in bar_list:
hash_check = sha3_256((bar.output_hash + bar.governing_receipt_id +
str(bar.turn_index)).encode())
assert hash_check == bar.content_hash
verify ML-DSA-65(bar.pqc_signature, bar.content_hash, pubkey)

Step 4 -- Verify genesis hash
expected_genesis = sha3_256((session_id + '||' + governing_receipt_id + '||' +
'OMNIX-CTCHC-GENESIS').encode())
assert expected_genesis == ctchc.genesis_hash
# Mismatch -> session identity tampered

Step 5 -- Reconstruct all chain links
prev = ctchc.genesis_hash
for bar in bar_list (ordered by turn_index):
expected_link = sha3_256(json.dumps({'prev':prev,'turn':bar.output_hash,'receipt':governin
g_receipt_id},sort_keys=True).encode())
assert expected_link == bar.chain_link # BEV-INV-011 + BEV-INV-018
prev = expected_link
assert prev == ctchc.final_chain_hash
# Mismatch -> substitution or reordering attack detected

Step 6 -- Report
return "VERIFIED" if all assertions passed else "NOT_VERIFIED: Step N failed [details]"
```

5.4 CTCHC Full Specification

```
CTCHC (Cross-Turn Coherence Hash Chain) -- identifier: CTCHC-{HEX16}

COMMITTED FIELDS:
ctchc_id : "CTCHC-9F3A2B1C4D5E6F7A"
session_id : "SESS-..." (UNIQUE -- one CTCHC per session)
governing_receipt_id : "ATFDR-..."
genesis_hash : "a3f2b1..." (SHA3-256 of session+receipt+GENESIS)
final_chain_hash : "c9d8e7..." (link[N-1], updated per turn)
turn_count : 7
all_bar_ids : ["BAR-...", "BAR-...", ...] (ordered by turn_index)
session_start_ns : 1748268000000000000
session_close_ns : 1748268120000000000
session_status : "COMPLETED" | "ACTIVE" | "HALTED" | "ABANDONED"
failure_turn_index : null | 4 (if halted)
failure_reason : null | "BEV-INV-003: bar_status=HALTED"

SEAL FIELDS:
seal_hash : SHA3-256(genesis || all_links_concat || final_chain_hash)
ctchc_seal : base64(ML-DSA-65.sign(seal_hash))
pqc_algorithm : "ML-DSA-65"
chain_sealed : true
atf_spec_version : "RFC-ATF-6_v1.0"
created_at : TIMESTAMPTZ
```

5.5 CTCHC Invariants: BEV-INV-010-014, BEV-INV-018

Invariant	Statement	Violation Consequence
BEV-INV-010	CTCHC genesis MUST be created before BAR_0. genesis = SHA3-256(session_id||"|"||receipt_id||"|"||"OMNIX-CTCHC-GENESIS").	Chain cannot be reconstructed. Session has no integrity anchor.
BEV-INV-011	link[n] = SHA3-256(json.dumps({prev, turn, receipt}, sort_keys=True)). Deterministic canonical form required.	Turn reordering, omission, substitution undetectable. SIP invalid.
BEV-INV-012	Verification MUST fail if any turn_index in [0, N-1] is absent. N turns claimed, N-1 links = REJECTED.	Partial session presented as complete. Forensic attribution incomplete.
BEV-INV-013	seal_hash = SHA3-256(genesis || all_link_hashes || tip_hash). Seal covers complete chain.	Partial seal accepted as complete. Tail turns unprotected.
BEV-INV-014	CTCHC seal MUST be ML-DSA-65 signed before OEP export or regulatory submission.	Session Integrity Proof not post-quantum secure. Regulatory submission blocked.
BEV-INV-018	Every chain link MUST carry same governing_receipt_id as CTCHC genesis. Prevents cross-session splicing.	Cross-session link splicing undetectable without chain recomputation.

5.6 CTCHC Database Schema

```
-- Table: atf_coherence_hash_chains (one row per session)
CREATE TABLE IF NOT EXISTS atf_coherence_hash_chains (
```

```
ctchc_id VARCHAR(64) PRIMARY KEY, -- CTCHC-{HEX16}
session_id VARCHAR(64) UNIQUE NOT NULL,
governing_receipt_id VARCHAR(128) NOT NULL,
genesis_hash VARCHAR(64) NOT NULL,
final_chain_hash VARCHAR(64),
turn_count INTEGER NOT NULL DEFAULT 0,
all_bar_ids JSONB,
session_start_ns BIGINT,
session_close_ns BIGINT,
session_status VARCHAR(32) NOT NULL DEFAULT 'ACTIVE'
CHECK (session_status IN ('ACTIVE','COMPLETED','HALTED','ABANDONED')),
failure_turn_index INTEGER,
failure_reason TEXT,
seal_hash VARCHAR(64),
ctchc_seal TEXT,
pqc_algorithm VARCHAR(16) NOT NULL DEFAULT 'ML-DSA-65',
chain_sealed BOOLEAN NOT NULL DEFAULT FALSE,
atf_spec_version VARCHAR(8) NOT NULL DEFAULT '1.6',
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

6. BEV Layer Composition

6.1 Per-Turn Execution Timeline

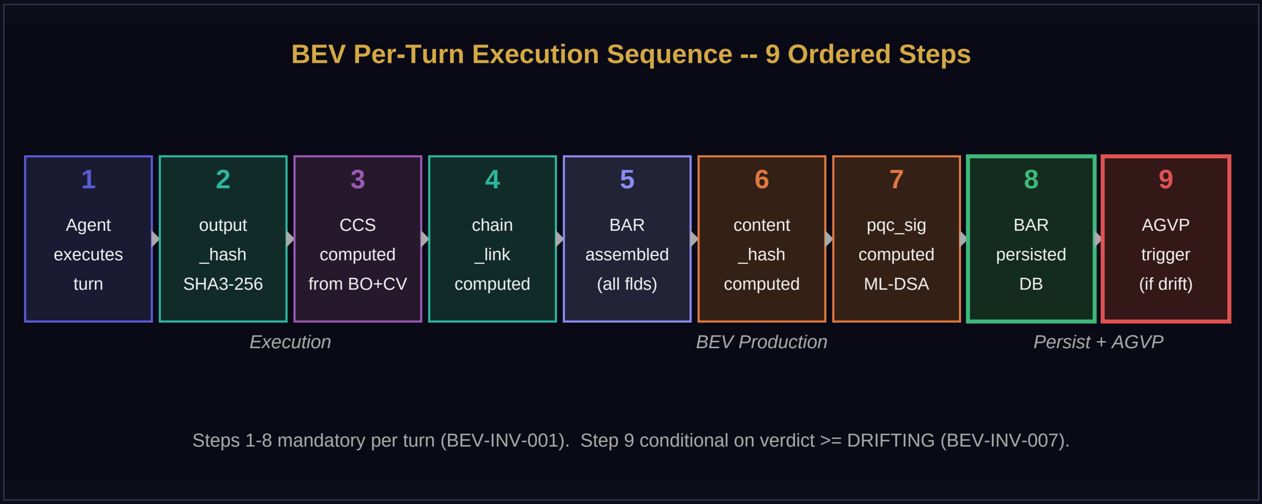

Figure 6. BEV per-turn execution timeline. Steps 1-8 mandatory for every turn (BEV-INV-001). Step 9 conditional on verdict (BEV-INV-007).

6.2 Three-Gap Closure

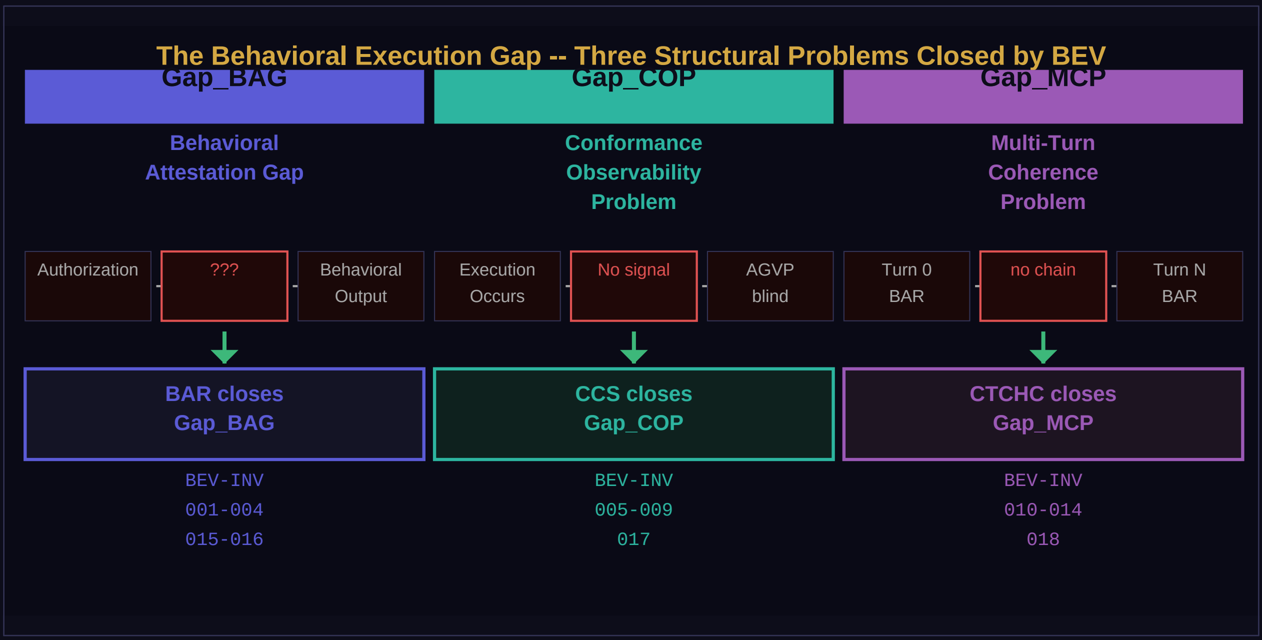

Figure 7. BEV closes the three structural behavioral governance gaps: BAR closes Gap_BAG, CCS closes Gap_COP, and CTCHC closes Gap_MCP.

6.3 Cross-Layer Integration Points

BEV component	Integrates with	RFC	Integration mechanism
BAR	Governing Receipt (L1-L2)	RFC-ATF-1	governing_receipt_id in every BAR; queryable from any ATF receipt

BEV component	Integrates with	RFC	Integration mechanism
CCS	AGVP Watchdog (L4)	RFC-ATF-4	DRIFTING/BREACH/VIOLATION -> PVR issuance via AGVP watchdog
CTCHC	OEP Forensic Package (L3)	RFC-ATF-3	CTCHC + BAR records included in OEP packages as SIP artifact
BAR	TCS Temporal Context (L5)	RFC-ATF-5	bar_timestamp_ns anchors to TCS for 7yr EU AI Act Art.72 retention
CCS	RAR Regulatory Projection (L5)	RFC-ATF-5	conformance history included in RAR field_projections for future review
CTCHC	GUGT Universal Invariant (L5)	RFC-ATF-5	SIP provides evidence for UGI-002 (offline-verifiable decision evidence)

6.4 Session Close Protocol

```
BEV SESSION CLOSE PROTOCOL (triggered on session end, HALT, or ABANDONED):

Step 11: Assert all turn_indices [0, N-1] present in BAR set (BEV-INV-012)
Step 12: Compute seal_hash = SHA3-256(genesis || all_links_concat || final_link)
Step 13: ctchc_seal = ML-DSA-65.sign(seal_hash) (BEV-INV-014)
Step 14: Update CTCHC record: chain_sealed=true, seal_hash, ctchc_seal
Step 15: Compute session_summary: bar_count, ccs_min, ccs_max, final_cumulative_drift
Step 16: If session was HALTED: record failure_turn_index + failure_reason
Step 17: Archive to OEP package if evidence lifecycle = HOT -> WARM threshold reached

CTCHC at close is the Session Integrity Proof (SIP):
Offline verifiable by any party holding: {ctchc.json, all_bars.json, public_key.pem}
No platform access required. No API credentials required. No network access required.
```

7. Formal Verification (OMNIX-FVS-1.0 Extension)

RFC-ATF-6 extends the OMNIX Formal Verification Suite (FVS-1.0) defined in RFC-ATF-4. RFC-ATF-6 maintains the dual methodology: Z3 SMT for arithmetic and structural invariants, TLA+ for state-machine safety and liveness. This is the only AI governance standard with machine-checkable formal proofs across both methodologies for behavioral execution verification.

7.1 Z3 SMT Proof Targets

Target ID	Module	Property Proven	Z3 Result
BEV-FVS-001	BAR	content_hash uniquely bound to (output_hash, receipt_id, turn_index) triplet	UNSAT
BEV-FVS-002	BAR	SHA3-256 structural uniqueness -- no two distinct inputs produce same content_hash	structural
BEV-FVS-003	CCS	conformance_score in [0.0, 1.0] for all valid component inputs	UNSAT
BEV-FVS-004	CCS	All four CCS components are non-negative for all valid inputs	UNSAT
BEV-FVS-005	CTCHC	Chain monotonicity: link[n] cryptographically depends on link[n-1]	structural
BEV-FVS-006	CCS	drift_delta = 1.0 - conformance_score is always non-negative	UNSAT
BEV-FVS-007	CCS	cumulative_drift is monotonically non-decreasing across turns	UNSAT
BEV-FVS-008	CTCHC	Missing turn_index in [0, N-1] causes verification = FAIL (gap attack)	UNSAT
BEV-FVS-009	BAR	BAR-HALT immediately disables subsequent turns (session state machine)	UNSAT
ATF-INV-001-BEV-EXT	Inherited	MAR invariant preserved under all BEV operations	UNSAT

7.2 BEV-FVS-003 -- CCS Score Bounds (Z3 proof)

```
from z3 import Real, Or, Solver, And, unsat

obs, css, sds, aas = [Real(x) for x in ['obs', 'css', 'sds', 'aas']]
solver = Solver()

# BEV-INV-006 component constraints
solver.add(obs >= 0, obs <= 40) # max(0, 40 - 10*violations) always in [0, 40]
solver.add(css >= 0, css <= 30) # max(0, 30 - 8*failures) always in [0, 30]
solver.add(sds >= 0, sds <= 20) # max(0, 20*(1-drift)) always in [0, 20]
solver.add(And(Or(aas == 0, aas == 10))) # binary authority check

ccs_score = obs + css + sds + aas # range: [0, 100]

# Claim: ccs_score/100 is ALWAYS in [0.0, 1.0]
# Try to find a violation:
solver.add(Or(ccs_score < 0, ccs_score > 100))
assert solver.check() == unsat # No violation exists -- BEV-INV-006 holds
print("BEV-FVS-003: UNSAT -- conformance_score in [0.0, 1.0] is guaranteed")
```

7.3 TLA+ Coverage

TLA+ Specification	Property Type	Coverage	Status
BEV_BAR.tla -- BAR_BEFORE_OUTPUT	Safety	BAR persisted before output delivery in all valid state transitions	PASS
BEV_BAR.tla -- HALT_PROPAGATES	Safety	HALTED BAR causes session state = HALTED in all reachable states	PASS
BEV_CCS.tla -- CCS_ATOMIC	Safety	CCS and BAR produced atomically -- no intermediate state with BAR but no CCS	PASS
BEV_CCS.tla -- AGVP_TRIGGERED	Safety	AGVP watchdog_triggered=true in all states with verdict in {DRIFTING,BREACH,VIOLATION}	PASS
BEV_CCS.tla -- DRIFT_MONOTONE	Safety	cumulative_drift is monotonically non-decreasing across all turn transitions	PASS
BEV_CTCHC.tla -- CHAIN_INIT_FIRST	Safety	genesis_hash set before first BAR in all valid session initialization sequences	PASS
BEV_CTCHC.tla -- SEAL_COMPLETENESS	Safety	chain_sealed=true only reachable when all turn_indices [0,N-1] present	PASS
BEV_INTEGRATION.tla -- BEV_NON_BLOCKING	Safety	CCS/CTCHC failure does not block primary record issuance (BAR-fail only)	PASS

Run all proofs: python omnix_core/formal_verification/run_bev_proofs.py

8. Combined Invariant Summary -- 106 Total

RFC-ATF-6 introduces 18 new invariants across three protocol families. Combined with the 88 invariants from RFC-ATF-1 through RFC-ATF-5, the complete ATF stack encompasses **106 formally specified invariants** across 20 protocol families.

Family	RFC	Count	Scope
ATF-INV	RFC-ATF-1	6	Identity & Delegation -- MAR, Acyclicity, Chain Root, Immutability, Non-Future-Dating, Offline Verifiability
RGC-INV	RFC-ATF-2	8	Runtime Continuity -- CES, AFG, HALT propagation, RC, RCR integrity
GPIL-INV	RFC-ATF-3	3	Governance Policy Interoperability Layer
ELR-INV	RFC-ATF-3	4	Evidence Lifecycle Record management
EAP-INV	RFC-ATF-3	7	Evidence Archive Pipeline
OEP-INV	RFC-ATF-3	6	OMNIX Evidence Package -- two-phase PQC seal, offline verification
FEA-INV	RFC-ATF-3	5	Forensic Export Authorization
FVP-INV	RFC-ATF-3	1	Forensic Verification Protocol
GECR-INV	RFC-ATF-3	6	Governance Execution Context Record
SGIP-INV	RFC-ATF-3	4	Semantic Governance Interoperability Protocol
DSPP-INV	RFC-ATF-4	7	Dynamic Semantic Portability Protocol -- TSA, SDR, RSA
AGV-INV	RFC-ATF-4	6	Anticipatory Governance Veto Protocol -- PVR
SSD-INV	RFC-ATF-4	3	Structural Shift Detection -- CRSI
FVS-INV	RFC-ATF-4	3	Formal Verification Suite extension
CGE-INV	RFC-ATF-5	7	Counterfactual Governance Engine
GUGT-INV	RFC-ATF-5	6	Grand Unified Governance Theory -- UGI-001-006
TGB-INV	RFC-ATF-5	5	Temporal Governance Bridge -- TCS, RAR, TMR
BEV-INV	RFC-ATF-6	18	Behavioral Execution Verification: BAR(001-004,015-016) + CCS(005-009,017) + CTCHC(010-014,018) [NEW]
TOTAL	--	106	Complete ATF Protocol Stack -- six RFCs -- 20 protocol families

9. ATF-BEV-Compliant Designation

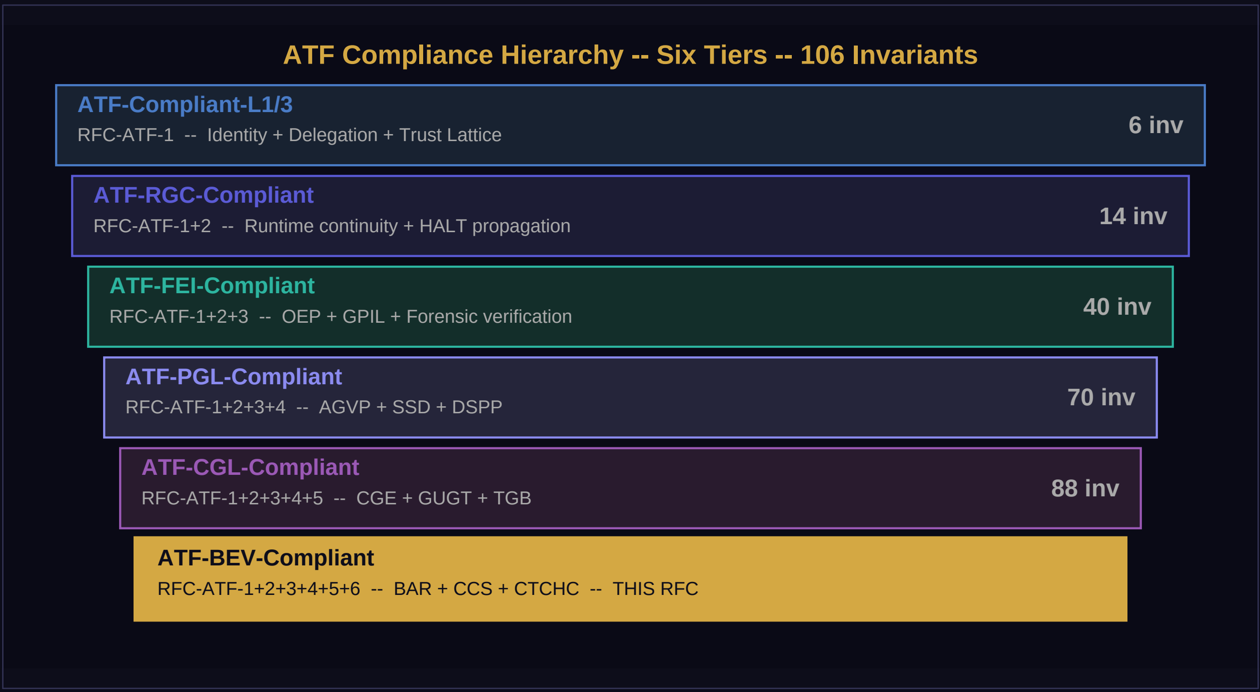

Figure 8. ATF compliance hierarchy. ATF-BEV-Compliant (gold) is the sixth and highest tier.

```
ATF COMPLIANCE HIERARCHY (six tiers, strictly hierarchical):

+-------------------------------------------------------------------+
| * ATF-BEV-Compliant (RFC-ATF-1+2+3+4+5+6) 106 inv HIGHEST |
| BAR + CCS + CTCHC operational; BEV formal verification passing |
+-------------------------------------------------------------------+
^--- extends without replacing
+-------------------------------------------------------------------+
| ATF-CGL-Compliant (RFC-ATF-1+2+3+4+5) 88 inv |
| CGE + GUGT UIR + TGB operational |
+-------------------------------------------------------------------+
^--- extends without replacing
+-------------------------------------------------------------------+
| ATF-PGL-Compliant (RFC-ATF-1+2+3+4) 70 inv |
| AGVP + SSD + DSPP operational |
+-------------------------------------------------------------------+
^--- extends without replacing
+-------------------------------------------------------------------+
| ATF-FEI-Compliant (RFC-ATF-1+2+3) 40 inv |
| OEP + GPIL + Forensic verification operational |
+-------------------------------------------------------------------+
^--- extends without replacing
+-------------------------------------------------------------------+
| ATF-RGC-Compliant (RFC-ATF-1+2) 14 inv |
| Runtime continuity + HALT propagation operational |
+-------------------------------------------------------------------+
```

```
^--- extends without replacing
+------------------------------------------------------------------+
| ATF-Compliant-L1/2/3 (RFC-ATF-1) 6 inv |
| Identity + Delegation + Trust Lattice operational |
+------------------------------------------------------------------+

ATF-BEV-Compliant requirements (all six required simultaneously):
(a) ATF-CGL-Compliant: all RFC-ATF-1/2/3/4/5 requirements satisfied
(b) BAR operational: BEV_ENABLED=true, all turns produce persisted BARs;
BEV-INV-001-004, 015-016 all satisfied; output delivery blocked without BAR
(c) CCS operational: CCS_ENABLED=true, AGVP integration active for DRIFTING+;
BEV-INV-005-009, 017 all satisfied; drift accumulator isolated per session
(d) CTCHC operational: CTCHC_ENABLED=true, all completed sessions sealed with SIP;
BEV-INV-010-014, 018 all satisfied; offline verification procedure tested
(e) BEV formal verification: run_bev_proofs.py -- all 9 Z3 targets UNSAT/structural;
all 8 TLA+ checks PASS; FVS runner output archived as compliance evidence
```

10. Implementation Reference

10.1 Reference Implementation Components

Component	Role	Public Interface
BAR Engine	Per-turn BAR production, content_hash, ML-DSA-65 signing, DB persist, BEV-INV-001-004 enforcement	/api/atf/bev/bar/{session_id}/{turn_index}
CCS Engine	CCS computation, 4-component scoring, verdict assignment, AGVP notification, BEV-INV-005-009,017	/api/atf/bev/ccs/{session_id}/{turn_index}
CTCHC Manager	Genesis init, per-turn link computation, session close sealing, SIP production, BEV-INV-010-014,018	/api/atf/bev/ctchc/{session_id}
BEV Session	Orchestrates BAR+CCS+CTCHC atomically per turn; enforces sequencing; manages session state machine	/api/atf/bev/session/{session_id}/turn
BAR Verifier	Offline BAR content_hash + ML-DSA-65 verification; no platform access required	python omnix_atf_verify.py --bar bar.json
CTCHC Verifier	Offline 6-step SIP verification: completeness, links, genesis, seal; detects all 4 attacks	python omnix_atf_verify.py --ctchc ctchc.json --bars bars.json
CCS Replayer	Offline CCS score recomputation from BAR payload + constraint vector	python omnix_bev_replay.py --bars bars.json --cv cv.json
BEV DB Manager	3 new tables + 4 indexes; append-only triggers; FK integrity; auto-created on first request	Auto-created via CREATE TABLE IF NOT EXISTS

10.2 API Endpoints

Endpoint	Method	Description	Auth
/api/atf/bev/session/start	POST	Initialize BEV session: create CTCHC genesis, return session_id + ctchc_id	B2B API key
/api/atf/bev/session/{sid}/turn	POST	Execute turn: produce BAR+CCS atomically, advance CTCHC, return bar_id	B2B API key
/api/atf/bev/session/{sid}/close	POST	Close session: seal CTCHC, produce SIP, return ctchc_id + seal_hash	B2B API key
/api/atf/bev/bar/{bar_id}	GET	Retrieve single BAR by ID	B2B API key
/api/atf/bev/session/{sid}/bars	GET	Retrieve all BARs for session (ordered by turn_index)	B2B API key
/api/atf/bev/ctchc/{session_id}	GET	Retrieve CTCHC for session (includes seal when closed)	B2B API key
/api/atf/bev/ctchc/{sid}/verify	POST	Online CTCHC verification (offline CLI also available)	B2B API key
/api/atf/bev/session/{sid}/ccs	GET	Retrieve CCS history for session (ordered by turn_index)	B2B API key
/api/atf/bev/session/{sid}/sip	GET	Export Session Integrity Proof bundle: ctchc + all_bars + public_key	B2B API key

10.3 Configuration Parameters

Parameter	Default	Range	Description
BEV_ENABLED	false	true/false	Master switch: enables BAR+CCS+CTCHC per turn
CCS_ENABLED	true	true/false	CCS computation per BAR (requires BEV_ENABLED)
CTCHC_ENABLED	true	true/false	Hash chain per session (requires BEV_ENABLED)
CCS_DRIFT_THRESHOLD	0.35	0.05-0.50	Cumulative drift threshold triggering HALT verdict
BAR_OUTPUT_MODE	HASHED	FULL/HASHED/REDACTED	Output payload storage mode (BEV-INV-002)
CCS_VERDICT_THRESHOLDS	90/70/50	configurable	CONFORMANT/DRIFTING/BREACH/VIOLATION score thresholds
BEV_AGVP_INTEGRATION	true	true/false	CCS verdict triggers AGVP PVR issuance (BEV-INV-007)
BEV_ASYNC_TIMEOUT_S	10	1-120	Timeout for async CCS/CTCHC operations
CTCHC_SEAL_ON_HALT	true	true/false	Seal CTCHC even on HALT (partial SIP)
BEV_OEP_AUTO_INCLUDE	true	true/false	Auto-include BAR+CTCHC in OEP packages (RFC-ATF-3)

11. Security Considerations

Threat	Attack Vector	BEV Mitigation	Residual Risk
BAR Omission (BEV-T-001)	Adversary presents subset of BARs, omitting turns with violations	CTCHC: missing turn_index in [0,N-1] -> verification=FAIL (BEV-INV-012)	NONE -- structural guarantee
Output Substitution (BEV-T-002)	Replace BAR_i output_hash with hash from compliant session	CTCHC link[n] includes turn output_hash -- substituted hash breaks chain (BEV-INV-011)	NONE -- SHA3-256 preimage resistance
Turn Reordering (BEV-T-003)	Present BARs in different order to hide violation sequence	chain_link = SHA3-256({prev, turn, receipt}) -- order-dependent; reordering breaks chain	NONE -- structural
Session Mixing (BEV-T-004)	Interleave BARs from compliant session with non-compliant session	BEV-INV-018: link.governing_receipt_id must match genesis -- cross-session links rejected	NONE -- receipt binding
CCS Score Inflation (BEV-T-005)	Override conformance_score to CONFORMANT without computing	BEV-INV-006: Z3 proof BEV-FVS-003 proves score in [0,1]; component constraints Z3-proved	LOW -- requires signing key compromise
CTCHC Partial Seal (BEV-T-006)	Seal chain over N-1 links, claim turn_count = N	BEV-INV-013: seal_hash includes all_links_concat -- partial seal produces wrong hash	NONE -- SHA3-256 collision resistance
BAR Replay Attack (BEV-T-007)	Reuse valid BAR from prior session in new session context	governing_receipt_id + session_id in content_hash bind BAR to specific session+receipt	NONE -- context binding
PQC Key Compromise (BEV-T-008)	Attacker compromises ML-DSA-65 signing key; forges BAR signatures	Key rotation via platform secret key management; prior BARs remain verifiable with original key	LOW -- requires key management breach
Conformance Drift Concealment (BEV-T-009)	Reset cumulative_drift by creating new session to avoid HALT	Each session has independent drift accumulator (BEV-INV-017); cross-session pattern detection via AGVP	LOW -- AGVP cross-session monitoring
Empty Output Bypass (BEV-T-010)	Submit empty output_text to produce trivially compliant BAR	BEV-INV-015: empty/whitespace output_text -> bar_status=VIOLATION; session halted	NONE -- explicit check

12. Novel Contributions

1. Behavioral Anchor Record (BAR) with PQC Seal

First governance artifact that cryptographically binds an AI agent's actual output to the governing receipt that authorized it, via a three-input SHA3-256 binding (output_hash || receipt_id || turn_index) signed with ML-DSA-65. Existing output logging systems (SageMaker Model Monitor, Azure AI Studio) capture outputs but do not bind them cryptographically to the specific authorization record. The BAR closes this binding gap.

2. Constraint Conformance Signal (CCS) -- Governance-Native Behavioral Metric

First behavioral conformance signal that references the actual Constraint Vector from a PQC-signed Governing Receipt as its measurement specification. Existing monitoring solutions (Arize Phoenix, WhyLabs, Evidently AI) monitor against separately configured policy files, creating a receipt-to-policy binding gap. CCS uses the receipt itself as the CV -- the authorization IS the measurement specification.

3. CCS-AGVP Behavioral Integration Loop

First integration of behavioral output conformance signals into a proactive veto protocol. RFC-ATF-4 AGVP covered structural (CES, fragmentation) and semantic (DSPP) signals. CCS completes the AGVP input space by adding behavioral conformance to the watchdog input stream. This enables anticipatory vetoes based on behavioral trajectory, not just structural health.

4. Cross-Turn Coherence Hash Chain (CTCHC) -- Session Integrity Proof

First session-level hash chain for multi-turn AI agent governance, seeded by the governing receipt ID and sealed with ML-DSA-65 as a Session Integrity Proof (SIP). Audit log hash chains exist in distributed systems; the architectural innovation is their linkage to governance receipts, enabling forensic verification of complete multi-turn behavioral sessions against their authorization records.

5. Offline-Verifiable Session Integrity Proof (SIP)

The CTCHC chain seal is the first AI governance artifact that proves the completeness, ordering, and receipt-binding of an entire multi-turn session, verifiable offline without network access or platform credentials. The 6-step verification procedure detects all four session manipulation attacks (omission, reordering, substitution, session-mixing).

6. ATF-BEV-Compliant -- Six-Tier Formally Verified Compliance Hierarchy

First six-tier AI governance compliance designation backed by 106 formally specified invariants across 20 protocol families, dual formal verification (Z3 + TLA+), and post-quantum cryptography. The BEV tier completes the ATF stack from identity through behavioral execution, closing all six identified gaps across the governance lifecycle.

13. Distinction from Prior Art

Feature	RFC-ATF-6 (BEV)	Arize / WhyLabs	SageMaker Monitor	Azure AI Studio	NIST AI RMF
Per-turn PQC-signed behavioral record (BAR)	[OK] ML-DSA-65 per turn	[NO]	[NO]	[NO]	[NO]
Binding to specific authorization receipt	[OK] content_hash triplet	[NO]	[NO]	[NO]	[NO]
Governance-native CCS (CV from signed receipt)	[OK] BEV-INV-005	[NO] -- separate policy	[NO]	[NO]	[NO]
AGVP behavioral integration (PVR for drift)	[OK] BEV-INV-007	[NO]	[NO]	[NO]	[NO]
Cross-turn session hash chain (CTCHC)	[OK] genesis + per-turn links	[NO]	[NO]	[NO]	[NO]
Offline-verifiable Session Integrity Proof (SIP)	[OK] 6-step procedure	[NO]	[NO]	[NO]	[NO]
Omission / reordering attack detection	[OK] CTCHC structure	[NO]	[NO]	[NO]	[NO]
Z3 + TLA+ formal proofs for behavioral invariants	[OK] 9 Z3 + 8 TLA+	[NO]	[NO]	[NO]	[NO]
Post-quantum cryptography (ML-DSA-65)	[OK] FIPS 204	[NO]	[NO]	[NO]	[NO]
Offline full-chain verification (no platform)	[OK] CLI + JSON only	[NO]	[NO]	[NO]	[NO]
6-tier compliance hierarchy (106 invariants)	[OK] ATF-BEV-Compliant	[NO]	[NO]	[NO]	Partial

14. Regulatory Alignment

Framework / Article	BAR Contribution	CCS Contribution	CTCHC Contribution
EU AI Act Art. 9 -- Risk Management	Per-turn output attestation binds execution to risk management receipt	Continuous monitoring within governance-specified boundaries (Art.9(7))	Session-level integrity proof for risk management documentation
EU AI Act Art. 12 -- Record-Keeping	Append-only per-turn behavioral record for mandatory technical file	Conformance history projection for multi-year record review	Complete session audit trail -- no gaps possible
EU AI Act Art. 14 -- Human Oversight	Auditable output record enables human review of actual outputs	CCS verdict provides quantitative signal for human review trigger	Session coherence evidence for oversight reporting
EU AI Act Art. 72 -- 7yr Retention	BAR records included in HOT/WARM/COLD lifecycle (TGB TMR)	CCS history archived with TCS for temporal projection via RAR	SIP preserved across evidence lifecycle transitions
NIST AI RMF MEASURE 2.6 -- Monitoring	Attestation of actual outputs per authorized action	Governance-native monitoring signal bound to receipt specification	Session coherence evidence for MEASURE 2.6 documentation
ISO/IEC 42001 Sec.8.4 -- Operations	Operational output record per Sec.8.4 controls	CCS is operational conformance measurement artifact	Turn-by-turn operational control record
MiFID II Art. 17 -- Algo Trading	Pre/post-output record for algorithmic trading actions	Conformance signal for trading output domain verification	Multi-turn session integrity for trading session audit
SOC 2 Type II CC7.2 -- Monitoring	Per-event behavioral monitoring evidence	Quantitative conformance evidence per CC7.2 monitoring controls	Session integrity evidence for change monitoring controls

15. Known Limitations and Open Questions

Limitation	Mitigation	ADR Reference	Status
CCS computation requires Constraint Vector (CV) from GR; CV quality depends on GR authoring quality	CV validation schema enforced at GR issuance; empty CV produces AAS=0 only (never score inflation)	ADR-182	MANAGED
SDS (semantic drift) component requires embedding model; embedding model change invalidates historical SDS scores	embedding_model_version field in CV; RAR projection handles model version transitions via TGB rulebook	ADR-182	OPEN
CTCHC per-turn link adds ~50ns latency per turn on standard hardware	SHA3-256 is hardware-accelerated on modern CPUs; CTCHC_ENABLED=false for latency-critical non-regulated deployments	ADR-183	MANAGED
BAR output_hash_mode=HASHED loses output content for forensic replay; mode=FULL raises GDPR concerns	Three-mode design (FULL/HASHED/REDACTED) addresses all deployment contexts; mode per deployment config	ADR-181	BY DESIGN
CCS drift_threshold is session-scoped; long-running sessions accumulate drift without reset mechanism	Session design should scope sessions to logical governance units; CTCHC naturally bounds session length	ADR-182	OPEN
Offline SIP verification requires all BAR records; partial export prevents verification	SIP export endpoint returns complete bundle; partial exports flagged as INCOMPLETE_SIP	ADR-183	MANAGED
Multi-instance deployments require shared signing key or BAR/CTCHC per-instance shard	Covered by existing OMNIX signing key management (ADR-166); no BEV-specific key management required	ADR-181/183	INHERITED
BEV does not cover tool calls or function invocations -- only text outputs	RFC-ATF-7 (proposed) may address structured action attestation; BEV covers LLM output domain completely	ADR-181	KNOWN GAP

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Appendix A -- BEV Wire Formats (JSON Schemas)

A.1 BAR Wire Format


```
{
"$schema": "https://omnixquantum.net/schemas/atf/bev/bar-v1.0.json",
"bar_id": "BAR-[16 uppercase hex]",
"session_id": "[session identifier]",
"governing_receipt_id": "[ATF governing receipt ID]",
"agent_id": "AID-[DOMAIN]-[16HEX]",
"turn_index": [integer >= 0, strictly monotone per session],
"output_hash": "[64-char hex SHA3-256 of output_text.encode('utf-8')]",
"output_hash_mode": "FULL" | "HASHED" | "REDACTED",
"output_payload": { ...structured output... }, // mode=FULL only, null otherwise
"constraint_vector": { ...CV from governing_receipt... },
"ccs_score": [decimal 0.0000 - 100.0000],
"ccs_verdict": "CONFORMANT" | "DRIFTING" | "BREACH" | "VIOLATION" | "HALT" | "NO_DATA",
"ccs_components": { "obs": [0-40], "css": [0-30], "sds": [0-20], "aas": [0 or 10] },
"chain_link": "[64-char hex SHA3-256 chain link for this turn]",
"genesis_hash": "[64-char hex] or null", // non-null for turn_index=0
"session_start_ns": [integer nanosecond timestamp] or null,
"bar_timestamp_ns": [integer nanosecond timestamp],
"issued_at": "[ISO8601 with nanoseconds and timezone]",
"content_hash": "[64-char hex SHA3-256(output_hash||receipt_id||str(turn_index))]",
"pqc_signature": "[base64 ML-DSA-65 signature over content_hash.encode('utf-8')]",
"pqc_algorithm": "ML-DSA-65",
"atf_spec_version": "RFC-ATF-6_v1.0",
"bar_status": "ACTIVE" | "HALTED" | "VIOLATION"
}
```


A.2 CCS Wire Format


```
{
"$schema": "https://omnixquantum.net/schemas/atf/bev/ccs-v1.0.json",
"ccs_id": "CCS-[16 uppercase hex]",
"bar_id": "BAR-[16 uppercase hex]",
"session_id": "[session identifier]",
"governing_receipt_id": "[ATF governing receipt ID]",
"turn_index": [integer >= 0],
"conformance_score": [decimal 0.0000 - 1.0000],
"verdict": "CONFORMANT"|"DRIFTING"|"BREACH"|"VIOLATION"|"HALT"|"NO_DATA",
"output_boundary_score": [decimal 0.000 - 40.000],
"constraint_satisfaction_score": [decimal 0.000 - 30.000],
"semantic_drift_score": [decimal 0.000 - 20.000],
"authority_alignment_score": 0.0 | 10.0,
"boundary_violation_count": [integer >= 0],
"constraint_failure_count": [integer >= 0],
```

```
"drift_magnitude": [decimal 0.0000 - 1.0000],
"cumulative_drift": [decimal >= 0.0],
"drift_delta": [decimal 0.0000 - 1.0000],
"watchdog_triggered": true | false,
"agvp_pvr_id": "PVR-[16HEX]" | null,
"halt_triggered": true | false,
"chain_link_hash": "[64-char hex]",
"computed_at_ns": [integer nanosecond timestamp]
}
```

A.3 CTCHC Wire Format

```
{
"$schema": "https://omnixquantum.net/schemas/atf/bev/ctchc-v1.0.json",
"ctchc_id": "CTCHC-[16 uppercase hex]",
"session_id": "[session identifier -- UNIQUE constraint]",
"governing_receipt_id": "[ATF governing receipt ID]",
"genesis_hash": "[64-char hex
SHA3-256(session_id||'||'||receipt_id||'||'||'OMNIX-CTCHC-GENESIS')]",
"final_chain_hash": "[64-char hex link[N-1]]" | null,
"turn_count": [integer >= 0],
"all_bar_ids": ["BAR-...", "BAR-...", ...], // ordered by turn_index
"session_start_ns": [integer nanosecond timestamp] | null,
"session_close_ns": [integer nanosecond timestamp] | null,
"session_status": "ACTIVE" | "COMPLETED" | "HALTED" | "ABANDONED",
"failure_turn_index": [integer] | null,
"failure_reason": "[string]" | null,
"seal_hash": "[64-char hex SHA3-256(genesis||all_links_concat||final_hash)]" | null,
"ctchc_seal": "[base64 ML-DSA-65 signature over seal_hash.encode()]" | null,
"pqc_algorithm": "ML-DSA-65",
"chain_sealed": true | false,
"atf_spec_version": "RFC-ATF-6_v1.0",
"created_at": "[ISO8601 timestamp]"
}
```

Appendix B -- Cross-Layer Integration Map

The following table provides the complete cross-layer integration points between BEV (Layer 6) and all prior ATF layers. BEV extends without modifying any prior layer.

BEV Component	Layer	Integrated With	RFC	Integration Field	Direction
BAR	L1	Agent Identity Record (AIR)	RFC-ATF-1	agent_id in BAR bound to AIR identity	BAR -> AIR
BAR	L1	Delegation Receipt (DR)	RFC-ATF-1	governing_receipt_id traces to DR chain_root_id	BAR -> DR
BAR	L2	Runtime Continuity Record (RCR)	RFC-ATF-2	bar_status=HALTED propagates via RCR HALT mechanism	BAR -> RCR
CCS	L3	OMNIX Evidence Package (OEP)	RFC-ATF-3	CCS records included as behavioral evidence in OEP bundle	CCS -> OEP
CTCHC	L3	OEP Forensic Package	RFC-ATF-3	CTCHC seal (SIP) is root artifact in OEP behavioral section	CTCHC -> OEP
CCS	L4	AGVP Watchdog	RFC-ATF-4	verdict in {DRIFTING,BREACH,VIOLATION} -> PVR issuance request	CCS -> AGVP
BAR	L5	Temporal Context Snapshot (TCS)	RFC-ATF-5	bar_timestamp_ns matches TCS.issued_at_ns for 7yr retention	BAR <-> TCS
CCS	L5	Regulatory Alignment Receipt (RAR)	RFC-ATF-5	CCS conformance history included in RAR field_projections	CCS -> RAR
CTCHC	L5	Universal Invariant Receipt (UIR)	RFC-ATF-5	SIP provides behavioral evidence for UGI-002 assertion	CTCHC -> UIR
BAR	L5	Counterfactual Attestation Token (CAT)	RFC-ATF-5	BAR records are behavioral ground truth for CGE fork comparison	BAR -> CAT

Appendix C -- ATF-BEV-Compliant Compliance Checklist

The following checklist provides the complete set of requirements for achieving ATF-BEV-Compliant designation. All items are required.

C.1 Prior RFC Compliance (ATF-CGL-Compliant required)

[ ] ATF-Compliant-L3: AIR, DR, TAR, Trust Lattice operational; ATF-INV-001-006 all satisfied

[ ] ATF-RGC-Compliant: RCR, CES, AFG, RC operational; RGC-INV-001-008 all satisfied

[ ] ATF-FEI-Compliant: OEP, GPIL, ELC, FVP operational; all L3 invariants satisfied

[ ] ATF-PGL-Compliant: AGVP, SSD, DSPP operational; all L4 invariants satisfied

[ ] ATF-CGL-Compliant: CGE, GUGT UIR, TGB operational; all L5 invariants satisfied

C.2 BAR Requirements

[ ] BEV_ENABLED=true in production configuration

[ ] BEV-INV-001: BAR produced before output delivery; no output path without BAR

[ ] BEV-INV-002: content_hash = SHA3-256(output_hash || receipt_id || str(turn_index))

[ ] BEV-INV-003: bar_status=HALTED -> immediate session halt; no subsequent turn permitted

[ ] BEV-INV-004: ML-DSA-65 seal over content_hash; append-only enforced on atf_behavioral_anchor_records

[ ] BEV-INV-015: empty output_text -> bar_status=VIOLATION enforced

[ ] BEV-INV-016: BAR identifier format BAR-{HEX16} enforced at generation

[ ] Z3 proof targets BEV-FVS-001, BEV-FVS-002, BEV-FVS-009: UNSAT/structural

C.3 CCS Requirements

[ ] CCS_ENABLED=true; CCS computed atomically with each BAR (BEV-INV-005)

[ ] BEV-INV-006: conformance_score in [0.0, 1.0]; all components >= 0; Z3 BEV-FVS-003: UNSAT

[ ] BEV-INV-007: verdict DRIFTING/BREACH/VIOLATION -> AGVP watchdog notified; PVR issued

[ ] BEV-INV-008: cumulative_drift > CCS_DRIFT_THRESHOLD -> HALT; threshold <= 0.50 in production

[ ] BEV-INV-009: append-only enforced on atf_constraint_conformance_signals

[ ] BEV-INV-017: cumulative_drift accumulator isolated per session_id; reload on restart

[ ] Z3 proof targets BEV-FVS-006, BEV-FVS-007: UNSAT

C.4 CTCHC Requirements

[ ] CTCHC_ENABLED=true; genesis hash created before BAR_0 (BEV-INV-010)

[ ] BEV-INV-011: chain link formula SHA3-256(json.dumps({prev,turn,receipt},sort_keys=True))

[ ] BEV-INV-012: gap detection operational; verification fails on missing turn_index

[ ] BEV-INV-013: seal_hash = SHA3-256(genesis || all_links_concat || final_hash)

[ ] BEV-INV-014: ML-DSA-65 seal applied before OEP export or regulatory submission

[ ] BEV-INV-018: all links carry same governing_receipt_id as genesis; cross-session splice rejected

[ ] Offline 6-step SIP verification procedure documented and tested

[ ] Z3 proof targets BEV-FVS-005, BEV-FVS-008: UNSAT/structural

C.5 Formal Verification

[ ] All 9 Z3 proof targets (BEV-FVS-001 through BEV-FVS-009): UNSAT or structural

[ ] All 8 TLA+ specification checks pass (BEV_BAR, BEV_CCS, BEV_CTCHC, BEV_INTEGRATION)

[ ] FVS runner output: python omnix_core/formal_verification/run_bev_proofs.py

[ ] FVS output archived with UIR as supporting compliance evidence

Author's Address

Harold Alberto Nunes Rodelo (Editor)

OMNIX QUANTUM LTD

71-75 Shelton Street, Covent Garden

London WC2H 9JQ, England

Email: standards@omnixquantum.com

Web: omnixquantum.net

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